MCU / 64MHz HSI + HSI48 USB

USB-C + ESD + VBUS detection

TWO PARALLEL POWERED CAN PORTS

12V local branch + reverse polarity + ideal-diode OR + TVS + 3V3

Input examples

Prototype: no Cyphal traffic; transceiver remains in standby.

No onboard 120R termination. Use an end plug at the final module.

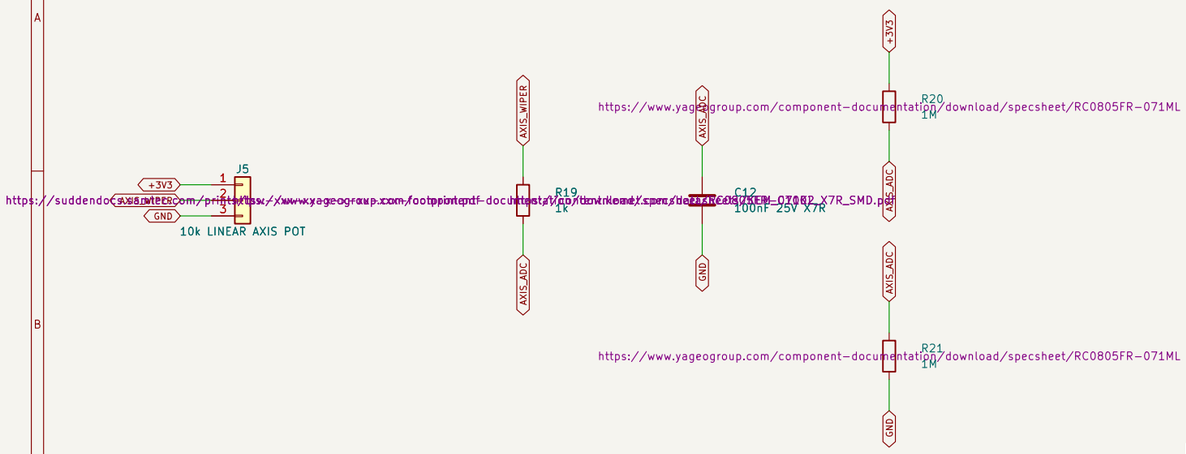
Connector pinout is provisional. NOT telephone/Ethernet compatible.

SWD and physical test input

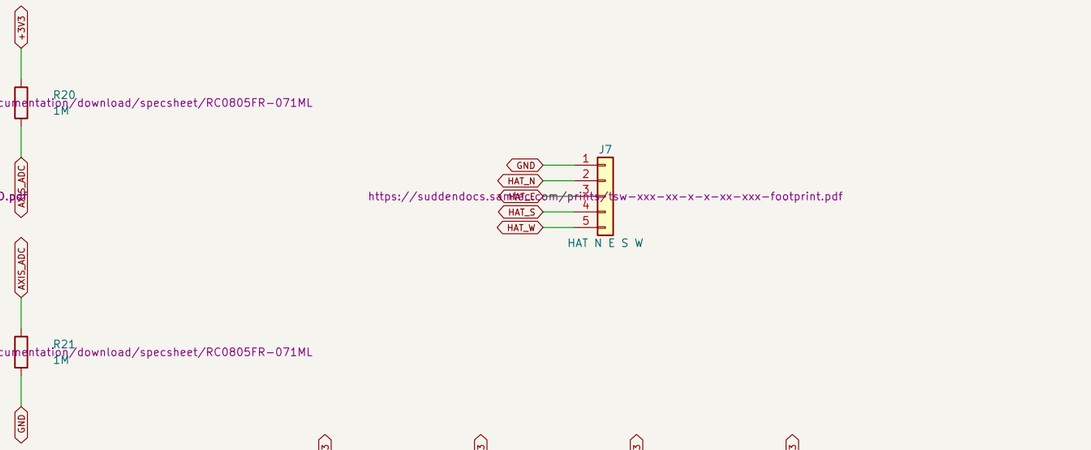
RJ12 pinout provisional; use qualified straight-through cable only
FABRICATION REVIEW — FOUR-LAYER + MILLING PROTOTYPE

OpenSimGear	
Sheet: /	
File: starter.kicad_sch	
Title: OpenSimGear STM32G0B1 starter	
Size: A2	Date: 2026-10-12
KiCad E.D.A. 10.0.6	**Rev: A draft**
	Id: 1/2

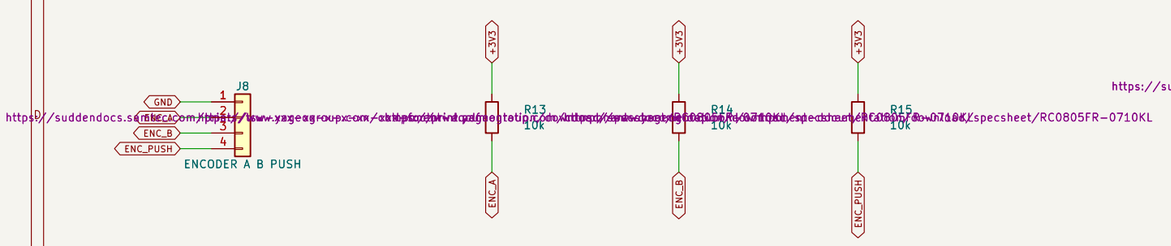
One linear axis: V+, wiper, ground. Bias centers an unplugged input.



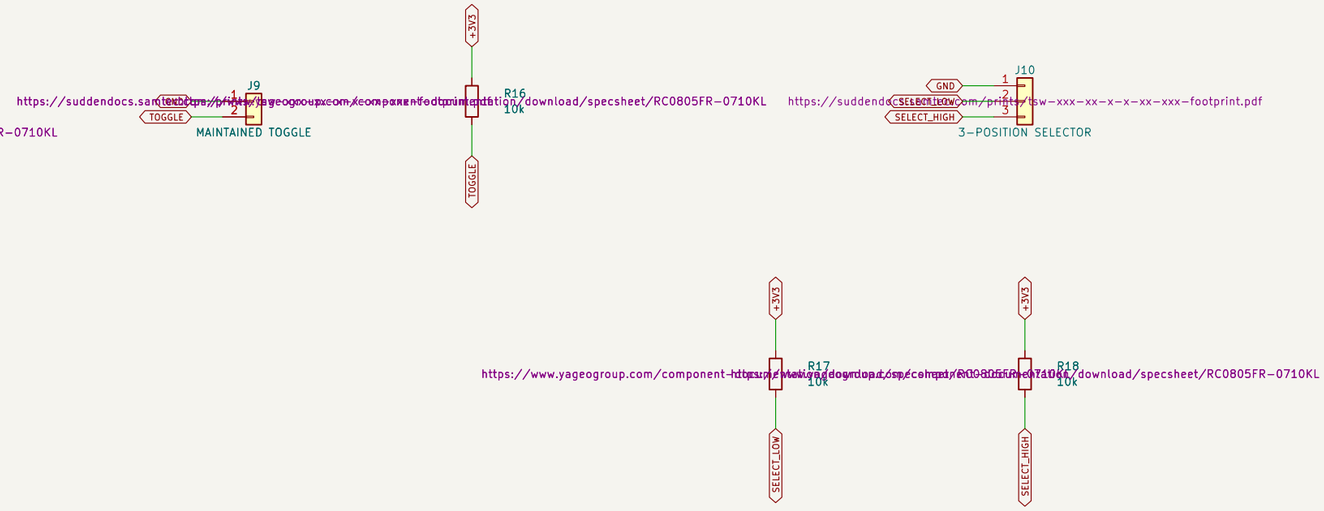
Four active-low contacts form one 8-way hat; adjacent pairs are diagonals.



One 4-transition-per-detent encoder, idle AB open; push is a separate button.



One maintained toggle and one center-open three-position selector.



External controls connect to labeled headers; 3.3V only

Sheet: /Input examples/ File: inputs.kicad_sch		
Title: OpenSimGear input examples		
Size: A3	Date: 2026-10-11	**Rev: A draft**
KiCad E.D.A. 10.0.6		Id: 2/2